

Digital VLSI Design

Project II: Design Of 32x32-Bit SRAM

Rupak Antani
2022102045

rupak.antani@students.iiit.ac.in

Prakhar Raj
2022102066

prakhar.raj@students.iiit.ac.in

Om Mehta
2022102046

om.mehta@students.iiit.ac.in

Himanshu Gupta
2022102002

himanshu.gupta@students.iiit.ac.in

I. INTRODUCTION

This report presents the design and implementation of a 32x32-bit static Random Access Memory (SRAM) block. The design includes important components such as a 5-bit adder, a 5-to-32 address decoder, an SRAM array, and a sense amplifier array. The project focuses on understanding how SRAM works and analyzing its performance through simulations.

Memory is a crucial part of digital systems for storing data, and this report is dedicated to designing a 32x32-bit SRAM for data storage. The SRAM array is built using 6T SRAM cells arranged in a 2D structure forming a 32x32 square array. Supporting components like the adder, address decoder, and sense amplifiers ensure the proper functioning of the array.

The functionality of this 1 KB SRAM array was verified using the Cadence Virtuoso tool with TSMC 180 nm technology.

The 32x32 SRAM design has the following features:

- **Data Storage and Retrieval** - It stores a 32-bit word across 32 rows, giving a total memory capacity of: $32 \times 32 = 1024$ bits (1 Kb).
- **Efficient Address Decoding** - The 5-to-32 decoder maps these address inputs to specific wordlines, enabling row-based access in the SRAM array.
- **Fast Read Operations** - Designed for quick read functionality, the circuit enables fast access to stored data. Also, a sense amplifier ensures that even small voltage changes in the SRAM cell during read operations are amplified to a usable logic level.
- **High-Speed Memory** - SRAM is typically used for cache memory or other high-speed applications due to its faster access times compared to other types of memory, such as DRAM.

II. TECHNOLOGY AND PARAMETERS

The following parameters were used for the design:

- **Technology node:** TSMC 180nm
- **Supply voltage:** 1.8 V
- **Threshold voltage of NMOS V_{thn} :** 0.50 V
- **Threshold voltage of PMOS V_{thp} :** -0.51 V

III. BLOCKS IN 32x32-BIT SRAM DESIGN

Designing 32x32-bit static-RAM block involves 4 major blocks at the circuit level as follows:

- **5-bit Adder:** Computes the effective address by using a 5-bit ripple carry adder.
- **5-32 Address decoder:** The address decoder takes in the 5 address lines $a4:0$ produced by the adder and decodes them to generate 32-word lines $wl31:0$ for the SRAM array.
- **SRAM Array:** Comprises of 32 rows and 32 columns, where each cell stores one bit.
- **Sense Amplifier Array:** The sense amplifiers at the end of the two complementary bit-lines amplify the small voltages to a normal logic level.

IV. SRAM CELL - WORKING & SIZING

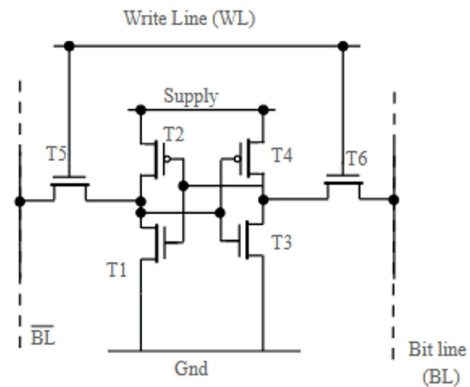


Fig. 1: 6T SRAM cell

SRAM retains data as long as power is supplied, with a 6-transistor cell storing one bit of information. The cell core consists of two CMOS inverters, where the output of one inverter feeds the input of the other, forming a feedback loop that stabilizes the state. Access transistors and word/bit lines (WL and BL) manage read/write operations.

Note that, the sizing of T5 and T6 transistors is 3μ and the sizing of the transistors T1, T2, T3 and T4 is 4μ .

In standby mode, the word line is low, deactivating the access transistors, while the inverters hold complementary values. To write data, bit lines impose data and its complement, $\overline{BL}$, then access transistors are activated by setting the word line high. Once the new data is stored, access transistors can be turned off, preserving the information in the inverters. For reading, the word line is activated, and the data is sensed on the bit lines.

SRAM Cell Design and Operation

a) **SRAM Design and Characteristics::** SRAM requires continuous power to retain its data, as it relies on a latch circuit. If the power is turned off, the data is lost. Unlike DRAM, SRAM does not use capacitors, so it avoids delays from charging, discharging, and refreshing. However, each SRAM cell needs six transistors to store a single bit, making it larger and more expensive than DRAM. Despite this, SRAM is preferred in cache memory and other high-speed applications because of its fast access times.

b) **Read Operation::** During a read operation, the SRAM cell's stored data is accessed without changing its state. First, the bitlines (BL and $\overline{BL}$) are pre-charged to the supply voltage (VDD). When the wordline (WL) is activated, it connects the stored data in the cell to the bitlines. This allows the cell's stored value to influence the bitlines. If the stored value in the SRAM cell is a '1', the bitline BL is pulled up towards VDD and $\overline{BL}$ is pulled down to 0 and if the value is a '0', the bitline $\overline{BL}$ is pulled towards VDD and BL is pulled towards 0. The bitline voltage changes are then sensed, allowing the data to be read without affecting the cell's state.

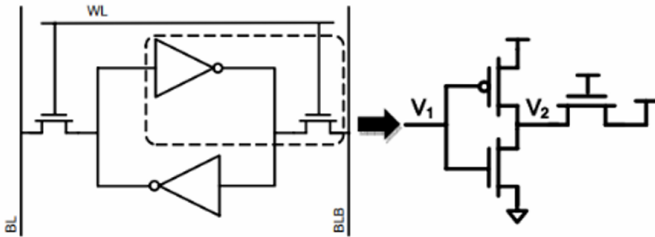


Fig. 2: SRAM read static noise margin

The SRAM cell resists significant voltage changes during the read operation, helping to maintain data stability. To evaluate this stability, the **Read Static Noise Margin (SNM)** is used, which is the largest square that can fit inside a butterfly plot—a graphical representation showing the relationship between stored data and bitline voltages. A larger SNM indicates that the cell can better resist disturbances during the read operation, making it more reliable.

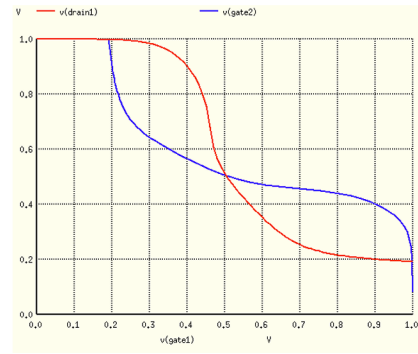


Fig. 3: Butterfly plot of SRAM Cell to find noise margins

c) **Write Operation:** The write operation in an SRAM cell involves changing the stored data by applying different voltage levels to the bitlines. In this operation, one bitline is pulled to 0V (logic '0'), while the other bitline is pulled to VDD (logic '1'). This voltage difference forces the cross-coupled inverters in the SRAM cell to flip the stored data. For example, if we want to write '0' to sram cell, $\overline{BL}$ is pulled to 1.8 V and BL is pulled to 0 V.

The **Write Noise Margin (WNM)** is a measure of how resistant the SRAM cell is to noise during the write operation. It is defined by the distance between the write voltage levels on the bitlines, and a larger WNM indicates that the cell is less likely to fail during a write. This margin ensures that the write operation is robust, even in the presence of noise or variations in voltage.

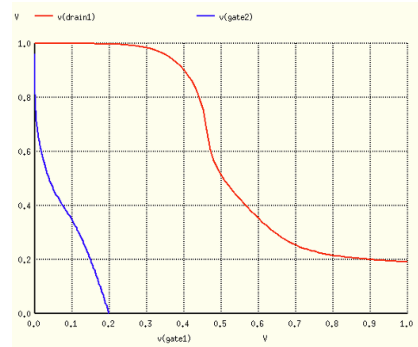


Fig. 4: Butterfly plot of SRAM Cell to find noise margins

These two operations, reading and writing, are fundamental to the functionality of an SRAM cell. The stability margins (SNM and WNM) play a critical role in ensuring that data is reliably read and written without corruption.

A. Transistor Sizing in a 6T SRAM Cell

The size of transistors in a 6T SRAM cell is crucial for proper read and write operations. Two main conditions determine the sizing:

- **Read Stability**

- During a read, the access transistors might disturb the stored value by pulling it towards the bitline, potentially flipping the data.
- To prevent this, the pull-down transistor must be stronger (wider) than the access transistors to ensure stability.

• Write Stability

- During a write, the access transistors must be strong enough to overpower the pull-up transistor to overwrite the stored value.
- This requires the access transistors to be wider than the pull-up transistors.

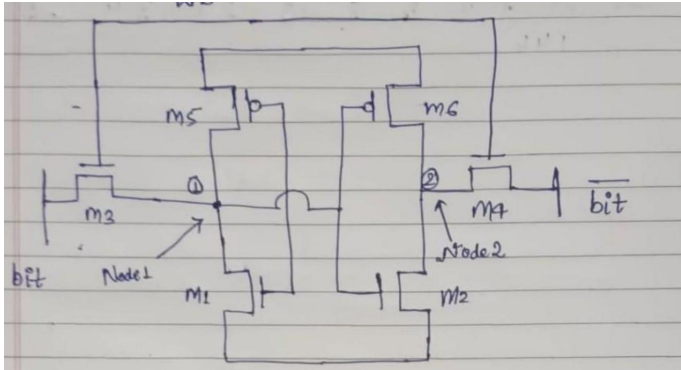
• Sizing Rules

- Read Stability: $W_{Pd} > W_{Pu}$
- Write Stability: $W_{Pu} > W_{Pl}$
- Retention Stability: W_{Pu} must counteract pull-down leakage.

In summary, the pull-down transistor is the widest, followed by the pull-up, with the access transistor being the narrowest.

Proof:

- Sizing in SRAM for Read and Write Operation



Read Operation:

- 1) Reading a '0' value at Node1 implies '1' is at Node2:

- a) bit, $\bar{\text{bit}} = V_{DD}$, $V_{\text{bit}} = N_{\bar{\text{bit}}} = V_{DD}$
- b) Wordline (WL) = VDD, $M_3 + M_4$ (ON)
- c) Node2 has '1' at inv1 input, so M_1 is ON.
- d) Node1 has '0' at inv2 input, so M_6 is ON.

- 2) Read and Write Stability criteria:

$$\left(\frac{\omega}{l}\right)_{\text{nmos}} > \left(\frac{\omega}{l}\right)_{\text{access}} > \left(\frac{\omega}{l}\right)_{\text{pmos}}$$

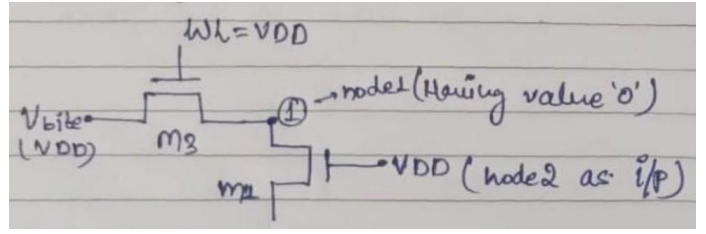
(Access - $m_3 + m_4$)

- 3) Now, looking into the M_3, M_1 circuit:

So,

$$V_{1,\text{max}} \leq V_{\text{th2}} \quad (V_{\text{th}} \text{ of node 2}) \quad (1)$$

and $\left(\frac{\omega}{L}\right)_{m1} > \left(\frac{\omega}{L}\right)_{m3}$ (criteria)



$$T1^{P(0)}(\text{Node } V_1) - V_{DD}(a) \rightarrow ON :$$

$$v_{S(0)} \left\{ \begin{array}{l} V_{DS} > V_{c,s} - V_{\tau,L} \\ s0, m_1 \rightarrow \text{Livear} \end{array} \right\}$$

All nmos have same V_{thn}

Now, m_3 (saturation), m_1 (Linear)

$$\frac{K_{n,m3}}{2} [v_{GS,m3} - V_{thn}]^2 = \frac{K_{n,m1}}{2} \left[2((v_{GS,m1} - v_{thn})v_{DS,m1}) - v_{DS,m1}^2 \right] \quad (1)$$

As,

$$m_3 : V_G = V_{DD} \quad \& \quad V_S = V_L$$

$$m_1 : V_G = V_{DD}; \quad V_D = V_{\perp}; \quad V_S = 0$$

Now,

$$V_1 \leq V_{T,n2} \quad (\because V_{T,n2} = V_{T,n}) \text{ Now,}$$

$$\Rightarrow V_1 = V_{T,n}$$

$$\frac{k_{n-1}}{2} [v_{DD} - v_1 - v_{1,n}]^2 = \frac{k_{n-1}}{2} [2(v_{DD} - v_{1,n})v_1 - v_1^2]$$

$$\text{as } V_1 = V_{T,h}$$

$$\frac{K_{n,m3}}{2} [v_{DD} - 2v_{t,h}]^2 = \frac{K_{n,m1}}{2} [2((v_{DD} - v_{t,n})v_{t,n} - v_{t,n}^2)]$$

$$\frac{k_{n3}}{k_{n1}} = \frac{2(V_{DD} - V_{t,n})V_{t,n} - V_{t,n}^2}{(V_{DD} - 2V_{t,n})^2}$$

$$\text{as } (\omega/L)m_3 < (\omega/L)m_1$$

$$\frac{(\omega/2)m_3}{(\omega/2)m_1} < 1$$

$$\Rightarrow \frac{(\omega/L)m_3}{(\omega/L)m_1} < \frac{2(V_{DD} - 1.5V_{t,n})V_{t,n}}{(V_{DD} - 2V_{t,n})^2}$$

Based on the theoretical calculation the value of the above ratios for read operation is obtained as: 0.8203 and based on the sizing taken by us in the circuit made in cadence virtuoso the ratio is: 0.75. Hence it satisfies the inequality and thus our sizing is correct.

Write Operation:

1) Assuming '1' is present at node1 implies '0' is present at node2.

a) Want to write '0' at node1:

i) Made bit = 0, $\overline{bit} = V_{DD}$ through write circuit ($V_{bit} = 0, V_{\overline{bit}} = V_{DD}$).

ii) Word Line (WL) = V_{DD} , M_3 & M_4 are 'on'.

iii) Since:

- Voltage at node1 is V_{DD} , so $M_1 \rightarrow$ 'ON'.
- Voltage at node2 is 0, so $M_5 \rightarrow$ 'ON'.

$\Rightarrow$ Now, looking into the circuit for further analysis.

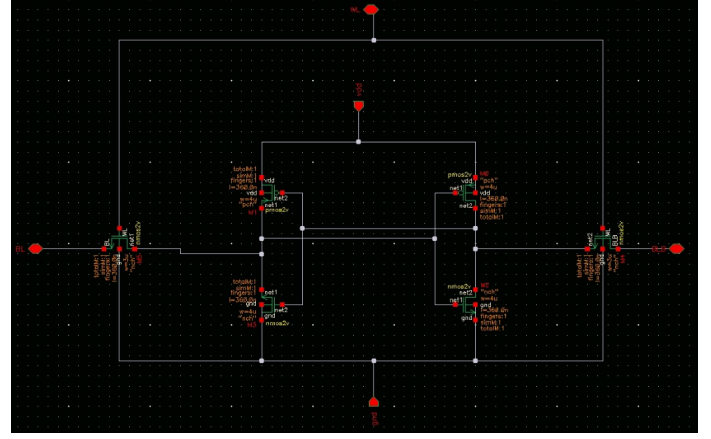
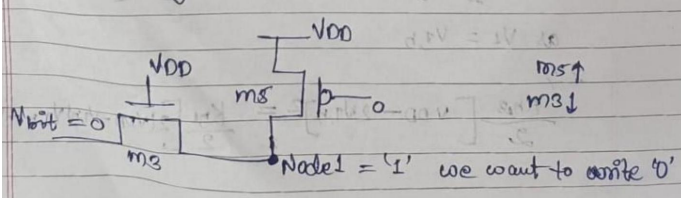


Fig. 5: SRAM Cell in Cadence Virtuoso



Now,

$$(\omega/L)m_3 > (\omega/L)m_5$$

$$\& V_1 \leq V_{tn2}$$

$$m_3 : V_{DS} \leq V_{GS} - V_{t,n} \text{ (linear), on}$$

$$m_5 : -V_{DS} \leq V_{GS} - V_{t,p} \text{ (saturation)}$$

Now, m_5 (saturation) & m_3 (linear)

$$\frac{K_{P,5}}{2} [V_{GS} - V_{t,p}]^2 = \frac{K_{P,3}}{2} [2(V_{GS} - V_{t,n})V_{DS} - V_{DS}^2]$$

As,

$$m_5 : v_G = 0, v_S = V_{DD}$$

$$m_3 : V_G = V_{DD}$$

$$V_s = V_{bit} = 0$$

$$V_D = V_1 \quad (\because V_1 = V_{t,n})$$

$$\frac{K_{P,5}}{2} [0 - v_{DD} - v_{t,p}]^2 = \frac{K_{P,3}}{2} [2(v_{DD} - v_{t,D})v_{t,n} - v_{t,n}^2]$$

$$\frac{K_{p,5}}{K_{p,3}} < \frac{2(V_{DD} - 1.5V_{t,n})v_{t,n}}{(V_{DD} + V_{t,p})^2}$$

$$\left(\because (w/l)_{m3} > (w/l)_{m5} \rightarrow \frac{(w/l)_{m3}}{(w/l)_{m5}} > 1 \right)$$

So, now,

$$\Rightarrow \frac{\left(\frac{w}{l}\right)_{m5}}{\left(\frac{w}{l}\right)_{m3}} < \frac{\mu_n \cdot 2(V_{DD} - 1.5V_{t,n})V_{t,n}}{\mu_P \cdot (V_{DD} + V_{t,p})^2}$$

Based on the theoretical calculation the value of the above ratios for read operation is obtained as: 3.06115 and based on the sizing taken by us in the circuit made in cadence virtuoso the ratio is: 1.33. Hence it satisfies the inequality and thus our sizing is correct.

V. 5-BIT ADDER

The 5-bit adder is a digital circuit used to add two 5-bit binary numbers. The 5-bit adder is designed by cascading five full adders, one for each bit, connected in a series. The output carry of each full adder is connected to the input carry of the next higher bit.

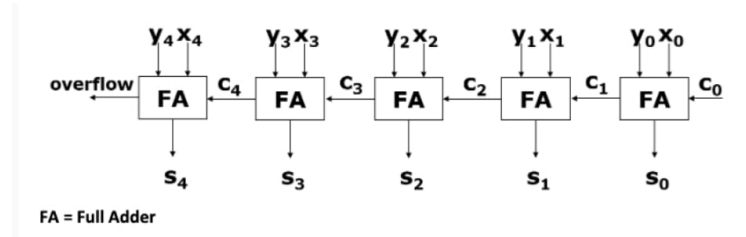


Fig. 6: 5-bit adder

A full-adder consists of logic gates, primarily XOR, AND, and OR gates, and that takes three inputs (two binary digits and a carry-in) and produces two outputs: a sum and a carry-out. It works by adding corresponding bits of the two binary numbers along with any carry from the previous bit.

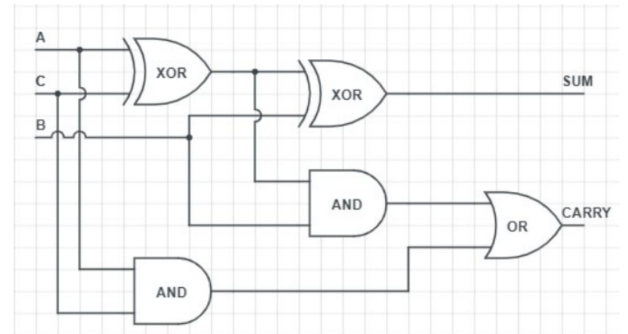


Fig. 7: Full adder Circuit

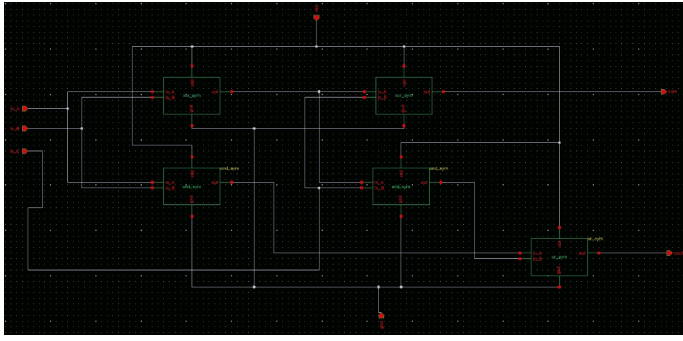


Fig. 8: Full adder in Cadence Virtuoso

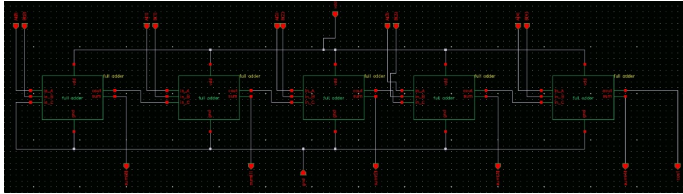


Fig. 9: 5-bit adder in Cadence Virtuoso

VI. 5-TO-32 DECODER

Address decoder is also known as row decoder. The row decoder used here is 5:32 decoder as there are 5 address lines. Since we are using 32x32 array there is a need of 32 word lines which is fulfilled by 5:32 decoder. The moment any one of the address line is selected, its corresponding word line goes high due to which the whole row gets selected. So this is the work of row decoder.

We are using five 3-to-8 decoder to build 5-to-32 decoder.

3 to 8 line decoder: The 3-to-8 line decoder is also known as Binary to Octal Decoder. In a 3 to 8 line decoder, there is a total of eight outputs, i.e., $Y_0, Y_1, Y_2, Y_3, Y_4, Y_5, Y_6, Y_7$ and three inputs, i.e., A_0, A_1, A_2 and one enable input 'E', as shown in Fig. 13. When enable 'E' is set to 1, one of these four outputs will be 1.

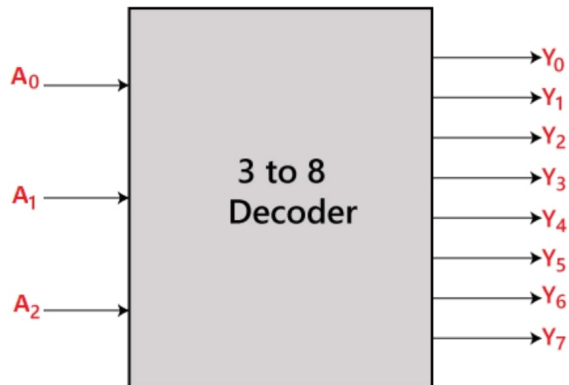


Fig. 10: 3-to-8 Decoder

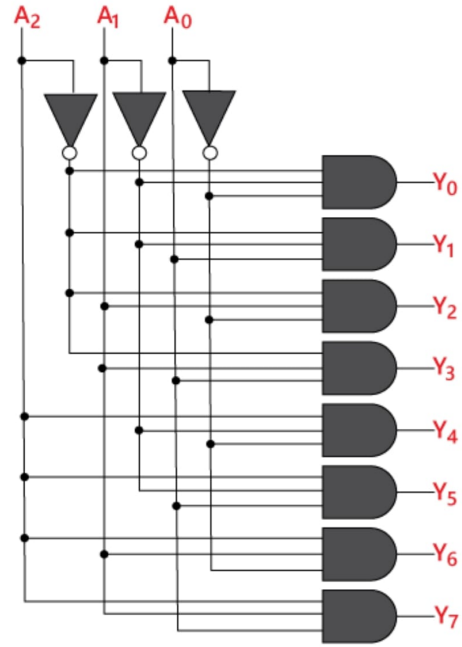


Fig. 11: 3-to-8 Decoder Circuit

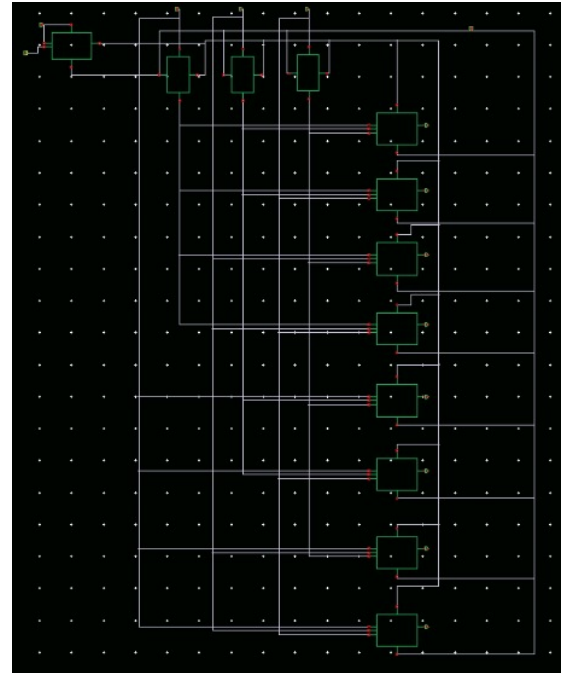


Fig. 12: 3-to-8 Decoder in Cadence Virtuoso

A 5-to-32 decoder is constructed using 3-to-8 decoders by employing a hierarchical approach. The 5-to-32 decoder takes 5 input lines and produces 32 outputs, where only one output is active (HIGH) at a time, corresponding to the binary value of the inputs.

Building the circuit: To build a 5-to-32 decoder using 3-to-8 decoders, we followed these steps:

Creating the First 3-to-8 Decoder: We built a 3-to-8 decoder to generate 8 outputs using the three least significant bits (LSBs) of the input. One of its inputs was connected to ground to fix its state.

Adding Four More 3-to-8 Decoders: We implemented four additional 3-to-8 decoders in parallel. These decoders were given the same three input lines. The “enable” inputs of these decoders were connected to the outputs of the first 3-to-8 decoder (one output per decoder).

Generating All 32 Outputs: Each of the four parallel 3-to-8 decoders generated 8 outputs, resulting in a total of 32 outputs.

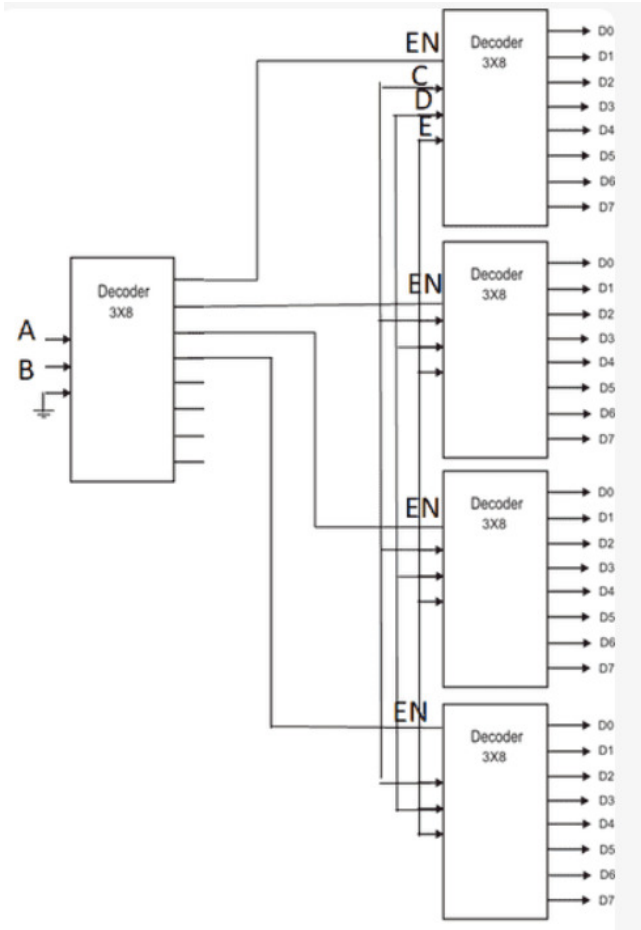


Fig. 13: 5-to-32 Decoder Circuit

VII. SRAM ARRAY

SRAMs are organized as an array of memory locations, where a memory access is either reading or writing all the bits in a single location. Here in Fig.18, we see the component layout for a 32-location SRAM array where each location hold 32 bits of data. The circuitry that we can see around the periphery is used to decode addresses and support read and write operations. To access the SRAM, we need provide enough address bits to uniquely specify the location.

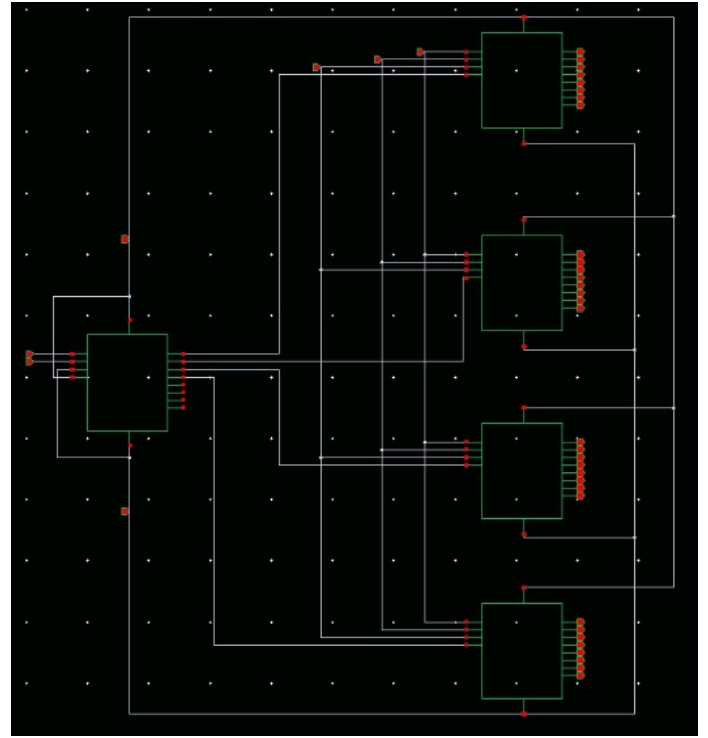


Fig. 14: 5-to-32 Decoder in Cadence Virtuoso

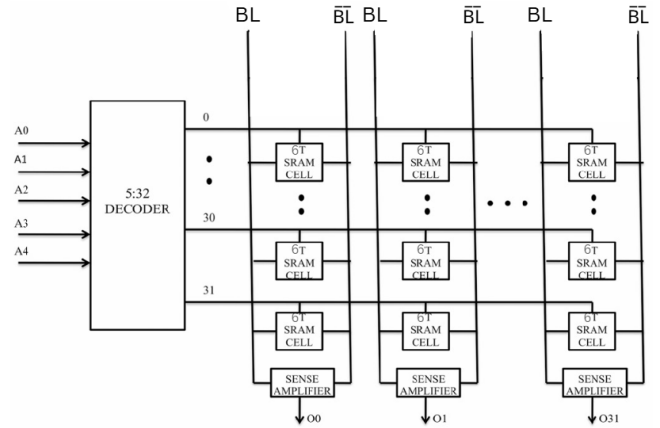


Fig. 15: SRAM Array Circuit

In this case we need 5 address bits to select one of the 32 memory locations. The address decoder logic sets one of the 32 wordlines (**the horizontal wires in the array**)(Fig.18) high to enable a particular row (location) for the upcoming access. The remaining wordlines are set low, disabling the cells they control. The active wordline enables each of the SRAM bit cells on the selected row, connecting each cell to a pair of bit lines (the vertical wires in the array i.e **BL & BL-bar**). During read operations the bit lines carry the analog signals from the enabled bit cells to the sense amplifiers, which convert the analog signals to digital data.

During write operations incoming data is driven onto the bit

lines to be stored into the enabled bit cells.

Precharging: Bitlines are precharged to VDD before reads to enhance signal stability.

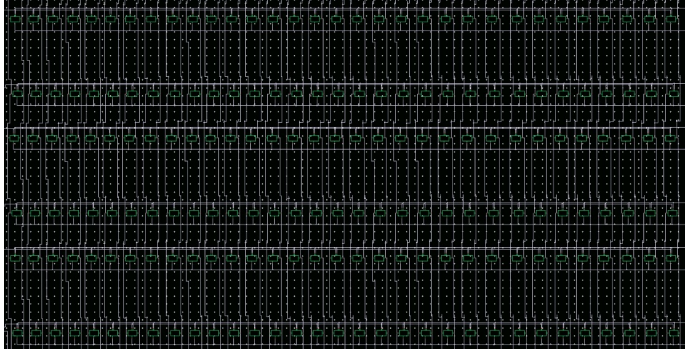


Fig. 16: SRAM Array in Cadence Virtuoso

Building the circuit:

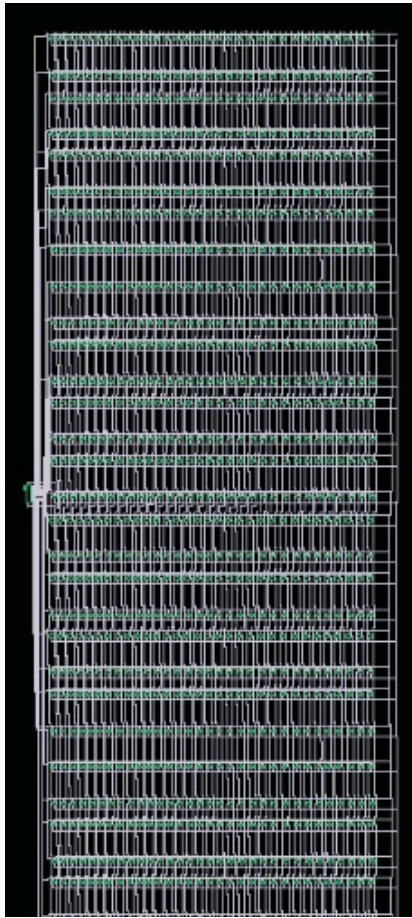


Fig. 18: 32x32 bit SRAM Design in Cadence Virtuoso

VIII. SENSE AMPLIFIER

The main purpose of sense amplifier is to measure the voltage difference between the two bit lines and as the name suggests amplify that difference i.e. amplifies the weak signal. The work of activating the sense amplifier is done by enabling

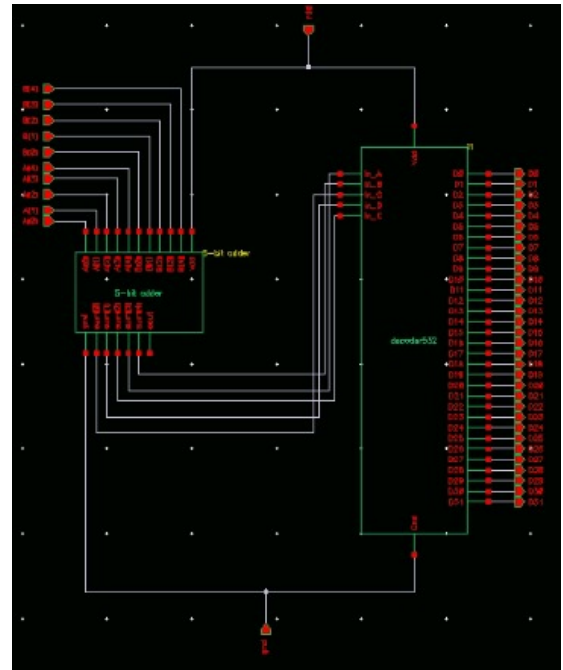


Fig. 17: Combined 5-bit adder and 5-to-32 decoder in Cadence Virtuoso

the sense enable (SE). The reason for using sense amplifiers in an array is because of discharging of one of the bit lines which it needs to boost during read operation.

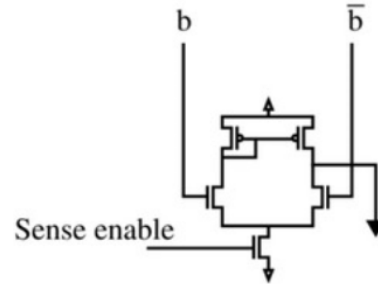


Fig. 19: Current Mirror Sense Amplifier Circuit
The sense amplifier detects and amplifies the small voltage changes during read operations.

The current mirror amplifier, commonly used as a sense amplifier in SRAM arrays. The differential signals from the bitlines serve as inputs to the sense amplifier circuit. Sense Enable signal activates the sense amplifier. When Sense Enable is asserted, it enables the current flow through the circuit. The circuit consists of MOSFETs arranged such that the current flowing through one branch (determined by the higher voltage bitline) is mirrored into the other branch. This amplifies the differential signal between b and $\bar{b}$. The sense amplifier converts the small voltage difference into a full-scale digital output.

The circuit quickly amplifies small bitline voltage differences, improving read operation speed. The current mirror design minimizes static power dissipation since the circuit is active only when Sense Enable is high. This circuit is ideal for SRAM applications due to its efficiency and ability to amplify small signals with minimal delay.

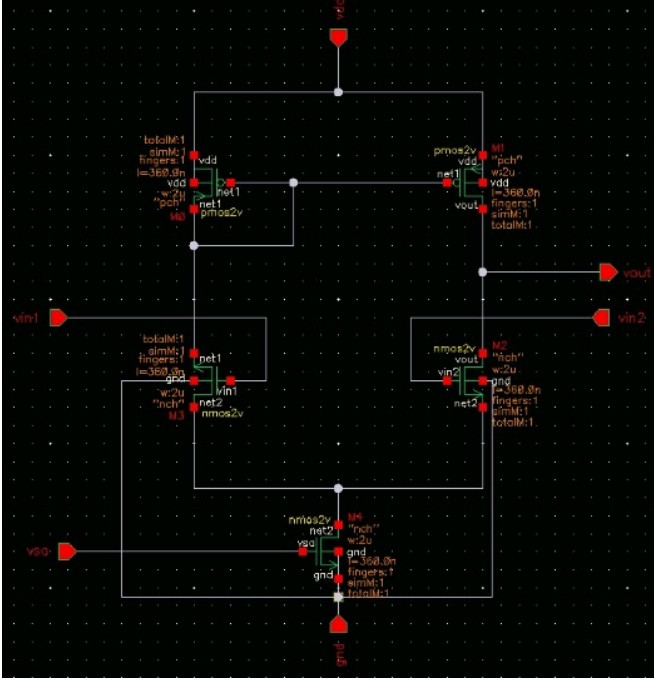


Fig. 20: Sense Amplifier in Cadence Virtuoso

IX. RESULTS & OBSERVATIONS

In [Fig. 21,22,23], X-axis denotes Voltage and & Y-axis denotes Time(in ns).

The Fig.21 shows the plot for reading an individual SRAM cell when the value stored in the cell is 1. We can observe that $\overline{bl}$ voltage decreases from an initial value of 1.8 to 0 V and bl voltage remains constant $\Rightarrow$. Hence, from this we can conclude that the value stored in the SRAM cell is '1'.

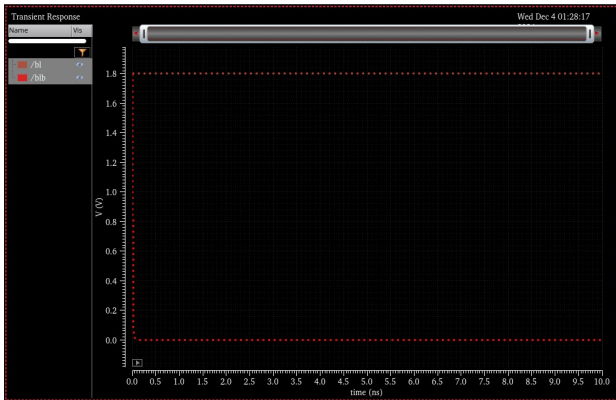


Fig. 21: Reading operation in SRAM Cell when stored value is 1 in Cadence Virtuoso

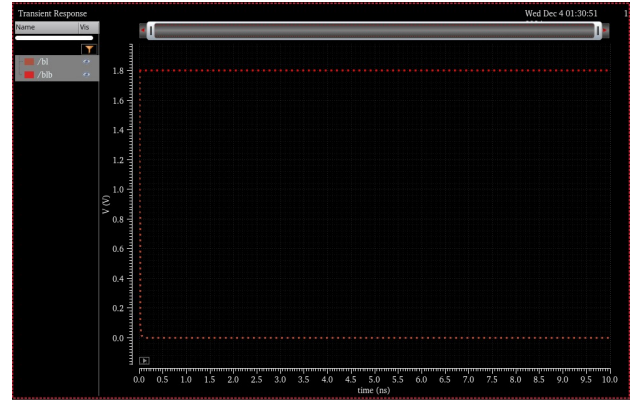


Fig. 22: Reading operation in SRAM Cell when stored value is 0 in Cadence Virtuoso

The Fig.22 shows the plot for reading an individual SRAM cell when the value stored in the cell is 0. We can observe that $\overline{bl}$ voltage decreases from an initial value of 1.8 to 0 V and bl voltage remains constant $\Rightarrow$ at 1.8 V. Hence, from this we can conclude that the value stored in the SRAM cell is '0'.

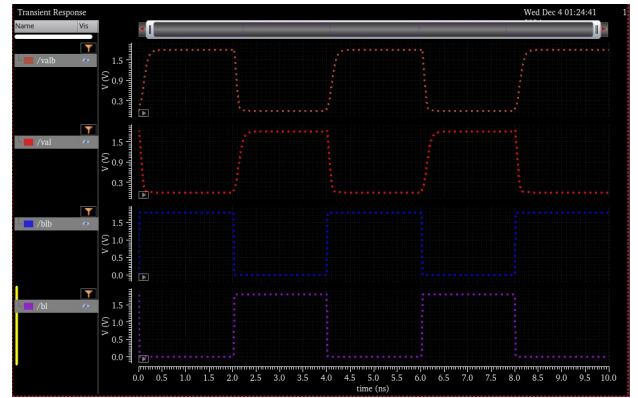


Fig. 23: Writing operation in SRAM Cell in Cadence Virtuoso BL('brown'), $\overline{BL}$ ('red'), Q2('pink'), Q1('orange')

Here, in 23, we are trying to verify our write operation. We are giving a pulse of logic low('0') & high('1') to Bit-lines('BL' & ' $\overline{BL}$ '). We can observe that the 'val', that is the value stored in the cell, follows the 'BL' or the bit-line values and the 'valb' follows ' $\overline{BL}$ ' values. Therefore, we can conclude that even the write operation works properly for our SRAM cell design. .

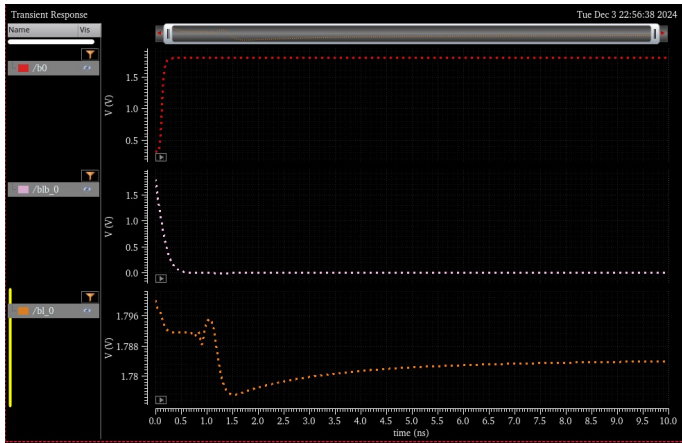


Fig. 24: Reading operation in combined SRAM design

The figure 24 shows the bit-line(bl_0), $\overline{\text{bit-line}}(blb_0)$ and the final value that is being read by the sense amplifier($b0$). It can be noticed that the blb_0 value decreases from the initial value of 1.8 to 0 V and the bl_0 value is almost constant at 1.8 V. Also, the final value of $b0$ is 1.8 V which shows that the logic value '1' is read and hence the value stored in the SRAM cell is 1.

While reading an individual SRAM cell, the delay is negligible as can be seen in Fig.21 and 22. However, when we try to read an sram cell in the combined circuit, the delay is close to 0.65 ns as can be seen in figure 24. This is due to the fact that the overall capacitance and resistance is much more in case of combined circuit compared to individual sram cell. Hence, while doing back to back reads, we need to wait for atleast 0.65 ns between consecutive read operations

X. CONTRIBUTIONS

- **Rupak Antani** - SRAM cell design, Logic Gates design, Combination of complete circuit, Report
- **Om Mehta** - Decoder design, Adder design and Report
- **Prakhar Raj** - Sense Amplifier design and Testing, Report
- **Himanshu Gupta** - Sense Amplifier design and Testing, Report